Agile Development Process and DevOps Lab
ISWE406P

Lab Midterm Examination

Date: 7th April, 2026

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23MIS0170

Question. To create a maven project for Student Attendance analysis and then use Jenkins to deploy a pipeline script.

File Edit Selection View Go Run Terminal Help — pom.xml - midterm - VSCodium

StudentAttendance.java | pom.xml | App.java

pom.xml

```
<project xmlns="http://maven.apache.org/POM/4.0.0"
         xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
         xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
         http://maven.apache.org/xsd/maven-4.0.0.xsd">

    <modelVersion>4.0.0</modelVersion>

    <groupId>com.example</groupId>
    <artifactId>attendance-app</artifactId>
    <version>1.0-SNAPSHOT</version>

    <properties>
        <maven.compiler.source>17</maven.compiler.source>
        <maven.compiler.target>17</maven.compiler.target>
    </properties>

</project>
```

EXPLORER: MIDTERM / src / main / java / com / example / App.java, StudentAttendance.java, test, pom.xml

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
sheheer@archlinux ~/Documents/Winter_25-26/agile/feb14/midterm git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: will change to "main" in Git 3.0. To configure the initial branch name
hint: to use in all of your new repositories, which will suppress this warning,
hint: call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
hint:
hint: Disable this message with "git config set advice.defaultBranchName false"
Initialized empty Git repository in /home/sheheer/Documents/Winter_25-26/agile/feb14/midterm/.git/
sheheer@archlinux ~/Documents/Winter_25-26/agile/feb14/midterm master ± git remote add origin https://github.com/mohammedsheheer/attendance-calculator.git
sheheer@archlinux ~/Documents/Winter_25-26/agile/feb14/midterm master ± git add .
sheheer@archlinux ~/Documents/Winter_25-26/agile/feb14/midterm master ± git commit -m "Initial Commit"
[master (root-commit) f5a7348] Initial Commit
 3 files changed, 75 insertions(+)
 create mode 100644 pom.xml
 create mode 100644 src/main/java/com/example/App.java
 create mode 100644 src/test/java/com/example/AppTest.java
sheheer@archlinux ~/Documents/Winter_25-26/agile/feb14/midterm master ± git add .
sheheer@archlinux ~/Documents/Winter_25-26/agile/feb14/midterm master ± git commit -m "Second Commit"
[master cd003c9] Second Commit
 3 files changed, 87 insertions(+), 64 deletions(-)
 create mode 100644 src/main/java/com/example/StudentAttendance.java
sheheer@archlinux ~/Documents/Winter_25-26/agile/feb14/midterm master
```

master 0↓ 1↑ ⊗ 0 ⚠ 0 — mohammedsheheer (now) Ln 17, Col 11 Spaces: 2 UTF-8 LF XML

File Edit Selection View Go Run Terminal Help — StudentAttendance.java - midterm - VSCodium

StudentAttendance.java | pom.xml | App.java

src > main > java > com > example > StudentAttendance.java

```
public class StudentAttendance {
    }

    public String getWarningSignal() {
        int percentage = getAttendancePercentage();
        if (percentage < 50) return "Critical Warning !!";
        if (percentage < 75) return "Low Attendance !!";
        return "Good Attendance, Well Done";
    }
}
```

EXPLORER: MIDTERM / src / main / java / com / example / App.java, StudentAttendance.java, test, target, pom.xml

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
sheheer@archlinux ~/Documents/Winter_25-26/agile/feb14/midterm master mvn compile
[INFO] Finished at: 2026-04-07T12:22:43+05:30
[INFO] ------------------------------------------------------------------------
sheheer@archlinux ~/Documents/Winter_25-26/agile/feb14/midterm master ± mvn exec:java -Dexec.mainClass="com.example.App"
[INFO] Scanning for projects...
[INFO]
[INFO] --------------------< com.example:attendance-app >--------------------
[INFO] Building attendance-app 1.0-SNAPSHOT
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- exec:3.6.3:java (default-cli) @ attendance-app ---
Enter student name: Johnny
Enter number of days: 5
Day 1 (1 = Present, 0 = Absent): 1
Day 2 (1 = Present, 0 = Absent): 1
Day 3 (1 = Present, 0 = Absent): 0
Day 4 (1 = Present, 0 = Absent): 0
Day 5 (1 = Present, 0 = Absent): 0

--- Report ---
Student: Johnny
Attendance: 40%
Eligibility: Not Eligible
Status: Critical Warning !!
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  13.752 s
[INFO] Finished at: 2026-04-07T12:23:04+05:30
[INFO] ------------------------------------------------------------------------
sheheer@archlinux ~/Documents/Winter_25-26/agile/feb14/midterm master ±
```

master* 0↓ 2↑ ⊗ 0 ⚠ 0 — mohammedsheheer (now) Ln 45, Col 43 Spaces: 4 UTF-8 LF Java

File Edit Selection View Go Run Terminal Help — StudentAttendance.java - midterm - VSCodium

StudentAttendance.java | pom.xml | App.java

src > main > java > com > example > StudentAttendance.java

```java
6   public class StudentAttendance {
39      }
40
41      public String getWarningSignal() {
42          int percentage = getAttendancePercentage();
43          if (percentage < 50) return "Critical Warning !!";
44          if (percentage < 75) return "Low Attendance !!";
45          return "Good Attendance, Well Done";
46      }
47  }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
sheheer@archlinux  ~/Documents/Winter_25-26/agile/feb14/midterm  master ±  mvn clean test
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/surefire/surefire-booter/3.5.4/surefire-booter-3.5.4.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/surefire/surefire-logger-api/3.5.4/surefire-logger-api-3.5.4.jar (14 kB at 25 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/surefire/surefire-extensions-spi/3.5.4/surefire-extensions-spi-3.5.4.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/surefire/surefire-extensions-api/3.5.4/surefire-extensions-api-3.5.4.jar (26 kB at 45 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/resolver/maven-resolver-util/1.4.1/maven-resolver-util-1.4.1.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/surefire/surefire-booter/3.5.4/surefire-booter-3.5.4.jar (119 kB at 163 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/resolver/maven-resolver-api/1.4.1/maven-resolver-api-1.4.1.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/surefire/maven-surefire-common/3.5.4/maven-surefire-common-3.5.4.jar (314 kB at 222 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/shared/maven-common-artifact-filters/3.4.0/maven-common-artifact-filters-3.4.0.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/resolver/maven-resolver-api/1.4.1/maven-resolver-api-1.4.1.jar (149 kB at 105 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-java/1.5.0/plexus-java-1.5.0.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/resolver/maven-resolver-util/1.4.1/maven-resolver-util-1.4.1.jar (168 kB at 102 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/ow2/asm/asm/9.8/asm-9.8.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/surefire/surefire-shared-utils/3.5.4/surefire-shared-utils-3.5.4.jar (2.9 MB at 1.7 MB/s)
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/surefire/surefire-extensions-spi/3.5.4/surefire-extensions-spi-3.5.4.jar (8.2 kB at 4.8 kB/s)
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/shared/maven-common-artifact-filters/3.4.0/maven-common-artifact-filters-3.4.0.jar (58 kB at 34 kB/s)
Downloaded from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-java/1.5.0/plexus-java-1.5.0.jar (57 kB at 33 kB/s)
Downloaded from central: https://repo.maven.apache.org/maven2/org/ow2/asm/asm/9.8/asm-9.8.jar (126 kB at 57 kB/s)
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  9.661 s
[INFO] Finished at: 2026-04-07T12:23:58+05:30
[INFO] ------------------------------------------------------------------------
sheheer@archlinux  ~/Documents/Winter_25-26/agile/feb14/midterm  master ±  git add .
sheheer@archlinux  ~/Documents/Winter_25-26/agile/feb14/midterm  master ±  git commit -m "Completed maven test commit"
[master 4cdc3e0] Completed maven test commit
 3 files changed, 1 insertion(+)
 create mode 100644 target/maven-status/maven-compiler-plugin/testCompile/default-testCompile/createdFiles.lst
 create mode 100644 target/maven-status/maven-compiler-plugin/testCompile/default-testCompile/inputFiles.lst
sheheer@archlinux  ~/Documents/Winter_25-26/agile/feb14/midterm  master
```

SOURCE CONTROL — REPOSITORIES: midterm mas... — CHANGES: Message (Ctrl+Enter to comm...) — Commit

G... Auto — Completed mave... master — Third Commit mohammedsheheer — Second Commit mohammedshe... — Initial Commit mohammedsheheer

master ⊗ 0 ⚠ 0 — mohammedsheheer (now) — Ln 45, Col 43 — Spaces: 4 — UTF-8 — LF — Java

File Edit Selection View Go Run Terminal Help — StudentAttendance.java - midterm - VSCodium

StudentAttendance.java | pom.xml | App.java

src > main > java > com > example > StudentAttendance.java

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45          return "Good Attendance, Well Done";
46      }
47  }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
* commit 4cdc3e0d351542e62849340dc1475fd602dd0ca5 (HEAD -> master, origin/master, origin/HEAD)
| Author: mohammedsheheer <mohammed.sheheer2@gmail.com>
| Date:   Tue Apr 7 12:24:55 2026 +0530
|
|     Completed maven test commit
|
* commit 086106e266d60c98a98bb522e17d9bee0d34d004
| Author: mohammedsheheer <mohammed.sheheer2@gmail.com>
| Date:   Tue Apr 7 12:22:35 2026 +0530
|
|     Third Commit
|
* commit cd003c911002416dd3a5d4f57a080573d39d8305
| Author: mohammedsheheer <mohammed.sheheer2@gmail.com>
| Date:   Tue Apr 7 12:19:51 2026 +0530
|
|     Second Commit
|
* commit f5a73487571e1781fcbc7950eec702b8d4aa06ac
  Author: mohammedsheheer <mohammed.sheheer2@gmail.com>
  Date:   Tue Apr 7 12:17:48 2026 +0530

      Initial Commit
(END)
```

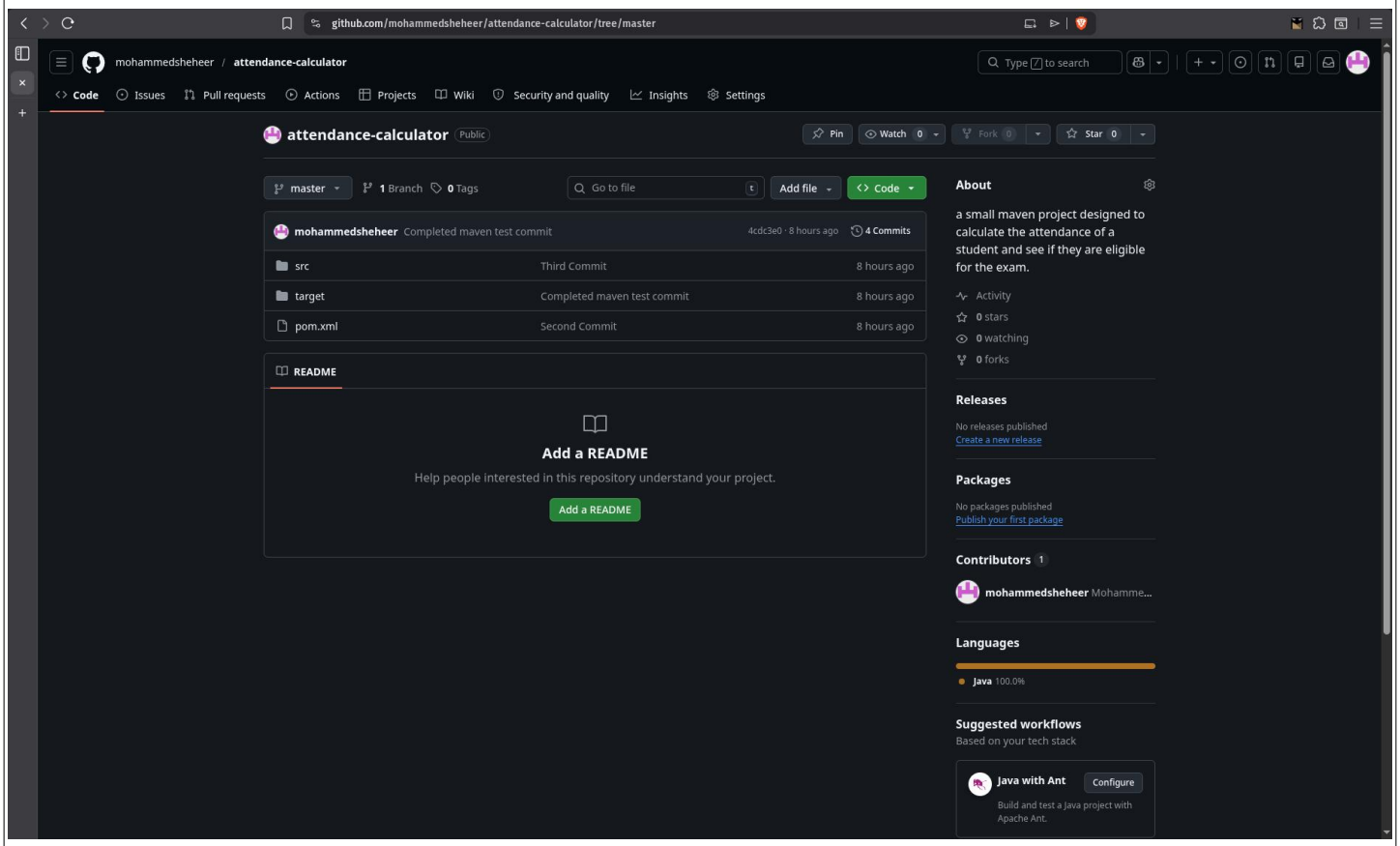
SOURCE CONTROL — REPOSITORIES: midterm mas... — CHANGES: Message (Ctrl+Enter to comm...) — Commit

G... Auto — Completed mave... master — Third Commit mohammedsheheer — Second Commit mohammedshe... — Initial Commit mohammedsheheer

master ⊗ 0 ⚠ 0 — mohammedsheheer (now) — Ln 45, Col 43 — Spaces: 4 — UTF-8 — LF — Java

File Structure(using `tree` command)

```
.
├── pom.xml
├── src
│   ├── main
│   │   └── java
│   │       └── com
│   │           └── example
│   │               ├── App.java
│   │               └── StudentAttendance.java
│   └── test
│       └── java
│           └── com
│               └── example
│                   └── AppTest.java
└── target
```



github.com/mohammedsheheer/attendance-calculator/tree/master/src

mohammedsheheer / attendance-calculator

Type / to search

Code | Issues | Pull requests | Actions | Projects | Wiki | Security and quality | Insights | Settings

master | attendance-calculator / src /

Go to file | Add file

mohammedsheheer Third Commit

086106e · 9 hours ago | History

Name	Last commit message	Last commit da...
..		
main/java/com/example	Third Commit	9 hours ago
test/java/com/example	Initial Commit	9 hours ago

github.com/mohammedsheheer/attendance-calculator/tree/master/src/main/java/com/example

mohammedsheheer / attendance-calculator

Type / to search

Code | Issues | Pull requests | Actions | Projects | Wiki | Security and quality | Insights | Settings

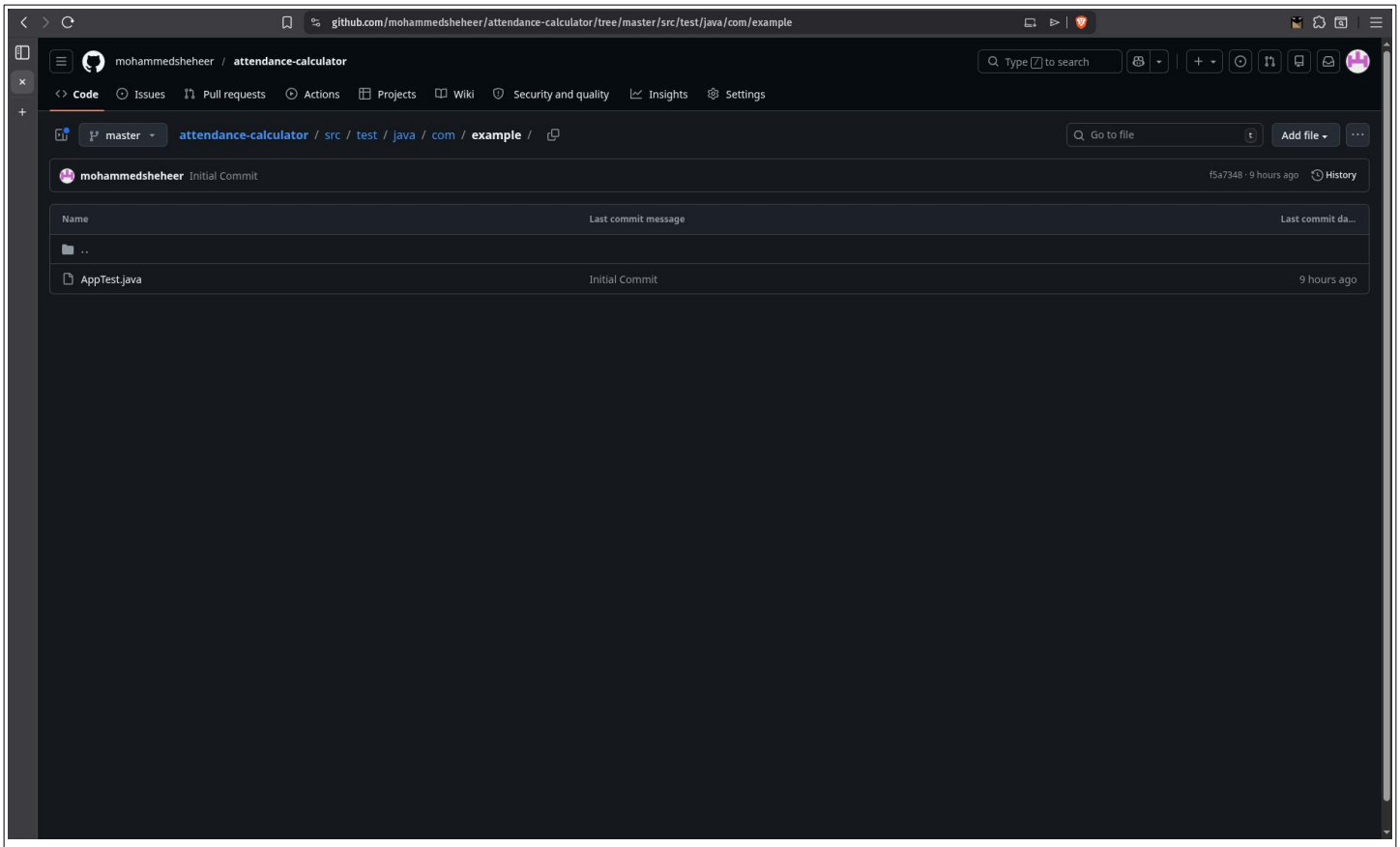
master | attendance-calculator / src / main / java / com / example /

Go to file | Add file

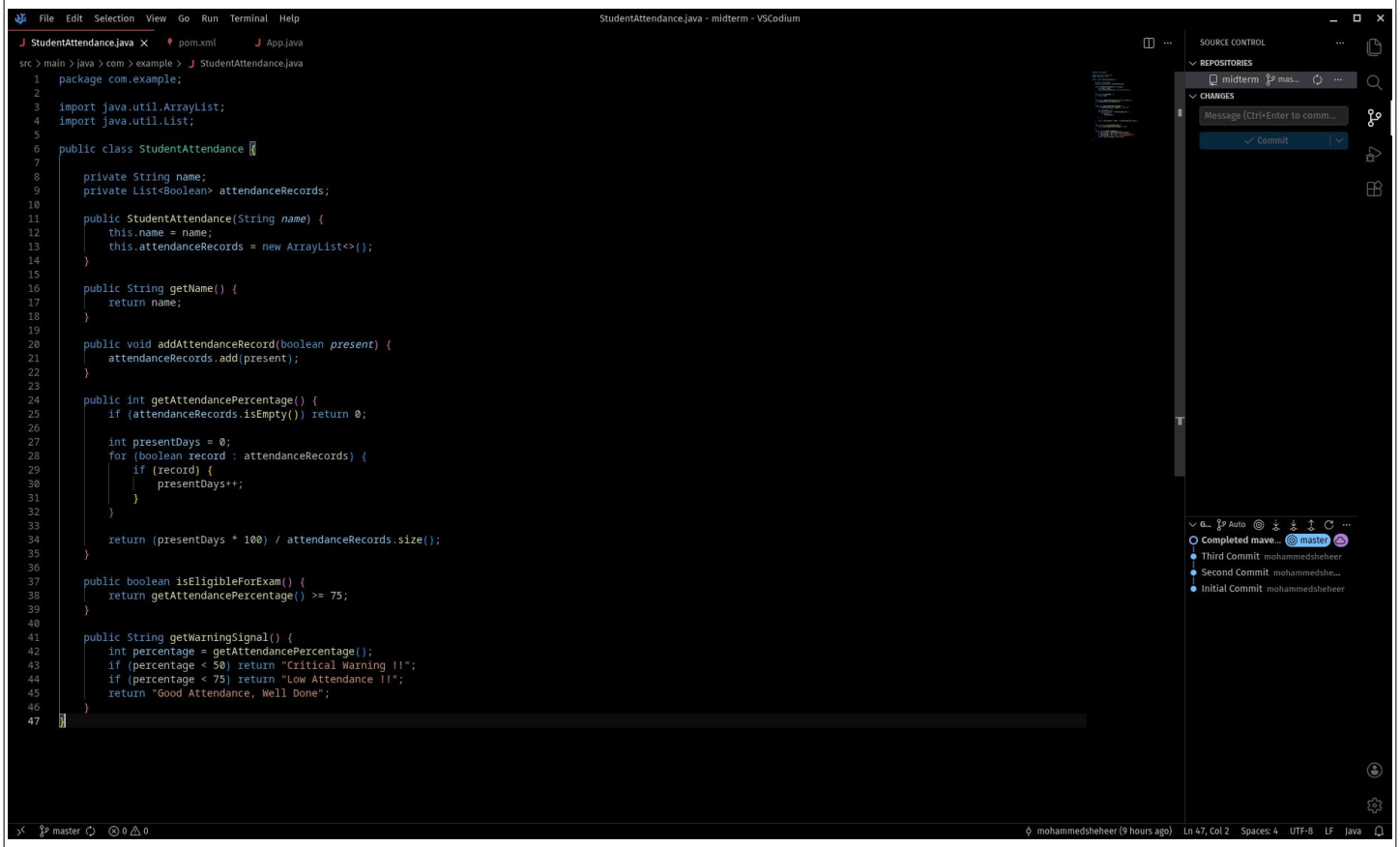
mohammedsheheer Third Commit

086106e · 9 hours ago | History

Name	Last commit message	Last commit da...
..		
App.java	Second Commit	9 hours ago
StudentAttendance.java	Third Commit	9 hours ago



StudentAttendance.java



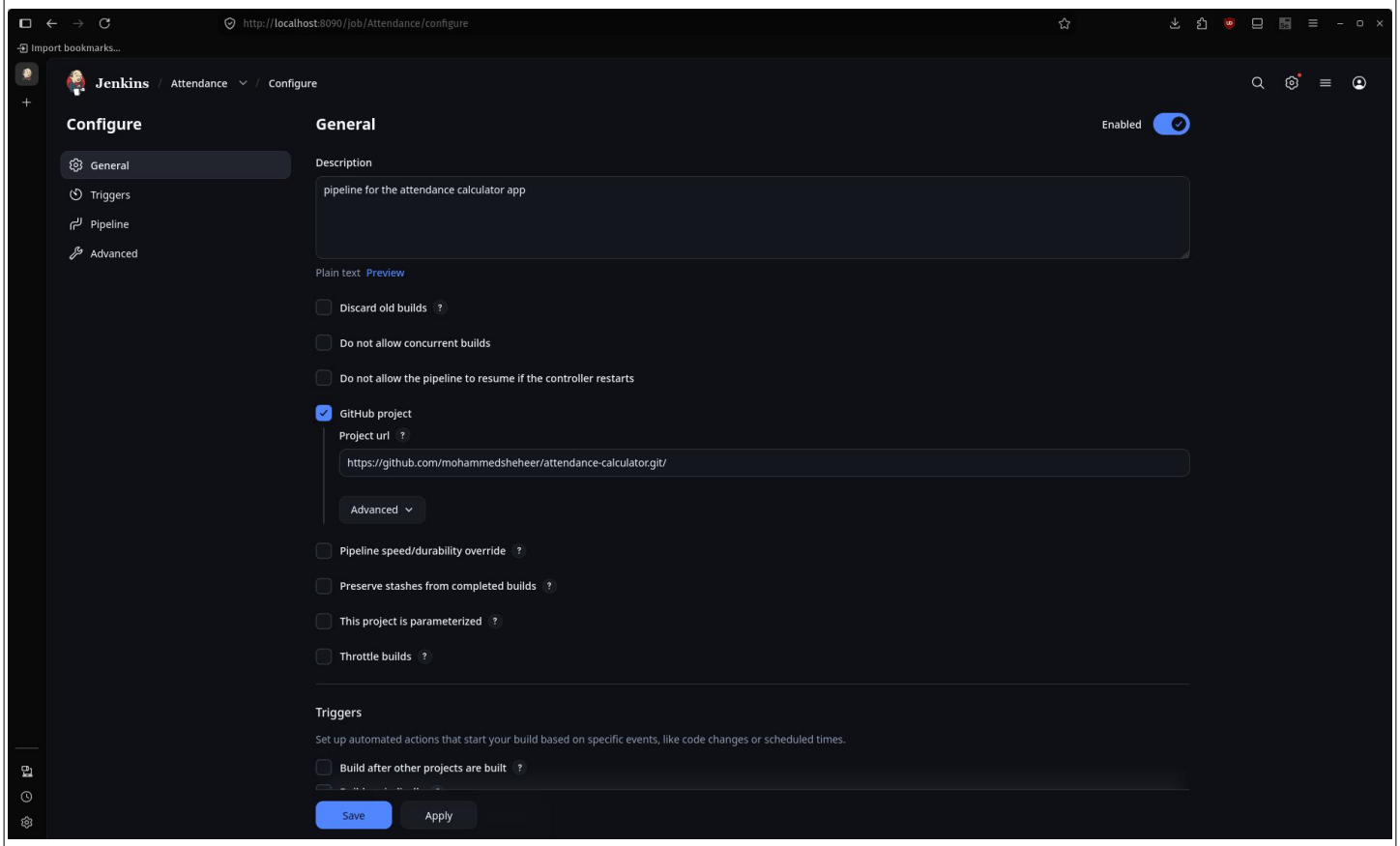
App.java

```java
package com.example;

import java.util.Scanner;

public class App {

    public static void main(String[] args) {

        Scanner scanner = new Scanner(System.in);

        System.out.print("Enter student name: ");
        String name = scanner.nextLine();

        StudentAttendance student = new StudentAttendance(name);

        System.out.print("Enter number of days: ");
        int numDays = scanner.nextInt();

        for (int i = 0; i < numDays; i++) {
            System.out.print("Day " + (i + 1) + " (1 = Present, 0 = Absent): ");
            int input = scanner.nextInt();
            student.addAttendanceRecord(input == 1);
        }

        System.out.println("\n--- Report ---");
        System.out.println("Student: " + student.getName());
        System.out.println("Attendance: " + student.getAttendancePercentage() + "%");
        System.out.println("Eligibility: " +
                (student.isEligibleForExam() ? "Eligible" : "Not Eligible"));
        System.out.println("Status: " + student.getWarningSignal());

        scanner.close();
    }
}
```

pom.xml

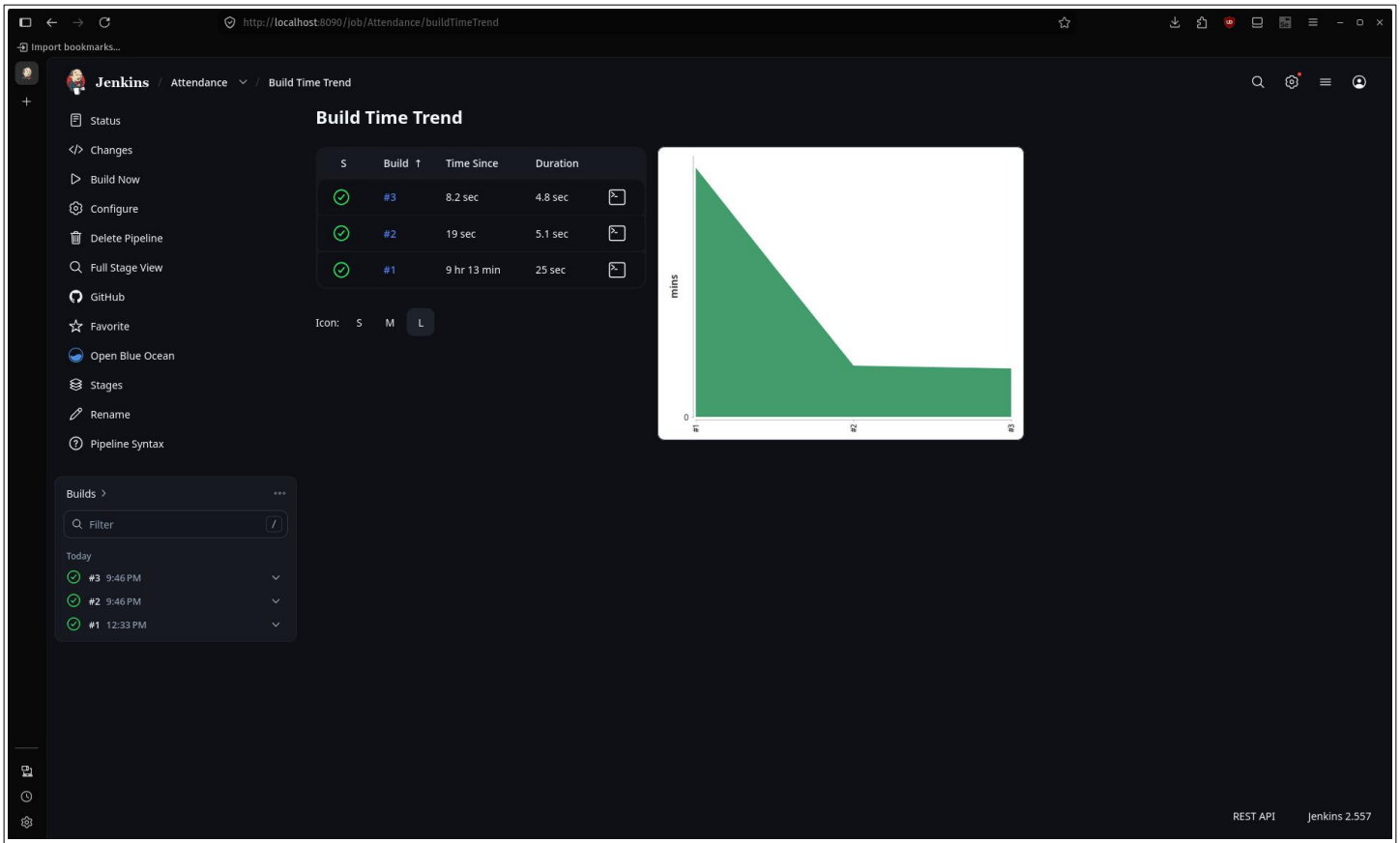
```xml
<project xmlns="http://maven.apache.org/POM/4.0.0"
         xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
         xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
         http://maven.apache.org/xsd/maven-4.0.0.xsd">

    <modelVersion>4.0.0</modelVersion>

    <groupId>com.example</groupId>
    <artifactId>attendance-app</artifactId>
    <version>1.0-SNAPSHOT</version>

    <properties>
        <maven.compiler.source>17</maven.compiler.source>
        <maven.compiler.target>17</maven.compiler.target>
    </properties>

</project>
```

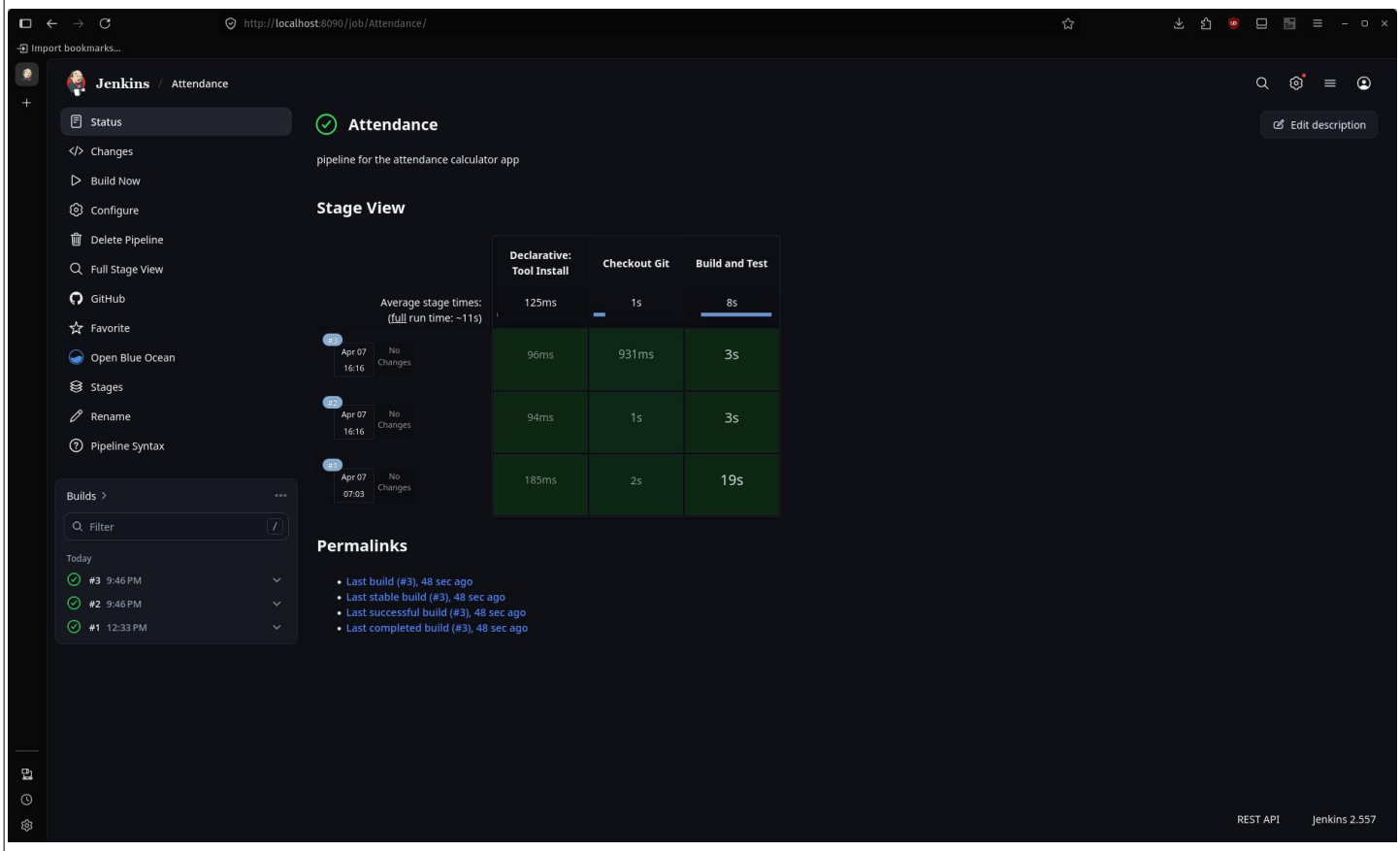
Jenkins Pipeline Configuration



Pipeline Script

```
pipeline {
    agent any
    tools {
        maven 'M3'
    }
    stages {
        stage('Checkout Git') {
            steps {
                git branch: 'master',
                url: 'https://github.com/mohammedsheheer/attendance-calculator.git'
            }
        }
        stage('Build and Test') {
            steps {
                sh 'mvn clean test'
            }
        }
    }
}
```


http://localhost:8090/job/Attendance/buildTimeTrend
Import bookmarks...
Jenkins / Attendance / Build Time Trend
Status
Changes
Build Now
Configure
Delete Pipeline
Full Stage View
GitHub
Favorite
Open Blue Ocean
Stages
Rename
Pipeline Syntax
Builds
Filter
Today
#3 9:46 PM
#2 9:46 PM
#1 12:33 PM
Build Time Trend
S
Build ↑
Time Since
Duration
#3 8.2 sec 4.8 sec
#2 19 sec 5.1 sec
#1 9 hr 13 min 25 sec
Icon: S M L
mins
0
#1
#2
#3
REST API
Jenkins 2.557


http://localhost:8090/job/Attendance/
Import bookmarks...
Jenkins / Attendance
Status
Changes
Build Now
Configure
Delete Pipeline
Full Stage View
GitHub
Favorite
Open Blue Ocean
Stages
Rename
Pipeline Syntax
Builds
Filter
Today
#3 9:46 PM
#2 9:46 PM
#1 12:33 PM
Attendance
Edit description
pipeline for the attendance calculator app
Stage View
Declarative: Tool Install
Checkout Git
Build and Test
Average stage times:
(full run time: ~11s)
125ms
1s
8s
#3 Apr 07 16:16 No Changes
96ms
931ms
3s
#2 Apr 07 16:16 No Changes
94ms
1s
3s
#1 Apr 07 07:03 No Changes
185ms
2s
19s
Permalinks
Last build (#3), 48 sec ago
Last stable build (#3), 48 sec ago
Last successful build (#3), 48 sec ago
Last completed build (#3), 48 sec ago
REST API
Jenkins 2.557